Vitor Pereira Matias

Website Scholar

Abstract

Currently pursuing a PhD degree in Computer Science with a focus on 3D reconstruction using Gaussian Splatting. I hold a Master's degree in computer vision focused on the classification of human skin tones, and a strong bachelor's degree in Pure Mathematics.

Studies

PhD Researcher — 06/2025 - current

State University of Sao Paulo (USP)

Emphasis in Gaussian Splatting, 3D and 4D reconstruction, specifically with a geometrically consistent reconstruction and implicit neural time flows.

Master's degree in Computer Science and Computational Mathematics — 03/2023 - 06/2025

State University of Sao Paulo (USP)

Emphasis in computer vision: skin tone classification of human faces in the wild using classic and deep learning approaches.

Bachelor's degree in Mathematics — 2018 - 2023

State University of Londrina (UEL)

Final Project: Covid-19 Prediction using compartmental models with optimization on real data.

Experience

PhD researcher — 11/2024 - current

USP and IMPA

Conducting research on 3DGS. We did a morphing of 3DGS representations. And we also devised a tutorial of 3DGS to showcase its power to the Brazilian CV community. Now focused on 4D and geometrically accurate reconstructions.

Junior Computer Vision Engineer — 03/2025 - 10/2025

Wikki/Wiise Brazil

Applied state of the art computer vision models (DINO) to microscopy slides of rocks to identify, segment, and count fossils that help dating the bedrocks.

Covid-19 Researcher — 2020 - 01/2024

Department of Mathematics State University of Londrina

As my mathematics bachelor's thesis. I researched the pandemic of Covid-19 for over two years by utilizing predictive mathematical models with optimization using Big Data analytics. For that i use the programming language Python with the common data science libraries such as NumPy, SciPy, Pandas, Scikit and Gurobi & others. Also, published a paper in 2021 and one to be published in 2025.

Education Tutoring Program (Programa de Educação Tutorial) — 04/2019 - 10/2022

Department of Mathematics State University of Londrina

In this program I've learned a lot of skills for over three years, such as project management with participation in over 40 unique projects. Also have developed abilities like team management, such as creating a new internal communication structure. Also as a software developer i have created LaTeX writing models used in writing three books and created two websites one for the group and other for an Event using HTML, CSS, JS. And one of the most important features that I learned was the process of writing a math book with over 300 pages with a team of 12 people.

Publications

Journals \Conferences:

- **Matias, V.**, Nascimento, F., Nakayama, K., Lima, J. P., Lobo, M., Wetzstein, G., Guibas, L., Paiva, A., and Novello, T. (2026). *PCAsplat: Gaussian splatting with local PCA regularization.* **arXiv**
https://arxiv.org/abs/2610.11011
- **Matias, V.**, Lobo, M., Neto, Novello, T., and J. B. (2026). *Large-scale dataset and benchmark for skin tone classification in the wild.* **arXiv**
https://arxiv.org/abs/2603.02475
- Bizzi, A., Grynberg, M., **Matias, V.**, Perazzo, D., Lima, J. P., Velho, L., and Novello, T. (2025). *Flowing: Implicit neural flows for structure-preserving morphing.* **NeurIPS**
https://arxiv.org/abs/2510.09537
- **Matias, V. P.**, Perazzo, D., Silva, V., Raposo, A., Velho, L., Paiva, A., Novello, T. 2025. *From Volume Rendering to 3D Gaussian Splatting: Theory and Applications.* **Main Track Tutorials** of the XXXVIII Conference on Graphics, Patterns and Images (SIBGRAPI).
https://arxiv.org/abs/2510.18101
- **Matias, V. P.**, Batista, J. B. 2024. *Enhancing Fairness in Machine Learning: Skin Tone Classification Using the Monk Skin Tone Scale.* Extended Annals of the XXXVII Conference on Graphics, Patterns and Images (SIBGRAPI).
DOI: 10.5753/sibgrapi.est.2024.31648, PDF: https://sol.sbc.org.br/index.php/sibgrapi_estendido/article/view/31648
- Cirilo, E. R., Natti, P. L., de Godoi, P. H. V., Lerma, A., **Matias, V. P.**,
& Romeiro, N. M. L. (2021). *COVID-19 in Londrina-pr: SEIR model with parameter optimization.* SEMINA. CIÊNCIAS EXATAS E TECNOLÓGICAS
DOI: https://doi.org/10.5433/1679-0375.2021v42n1Suplp45

Book/Chapters:

- **MATIAS, V. P.***; Et al*. . Matemática: um livro feito por alunos para alunos. 1. ed. Londrina: Pet Matemática UEL, 2022. v. 1. 342p .

* denotes equal contribution.
Check my scholar for more info https://scholar.google.com/citations?user=OWAq7C8AAAAJ&hl=en&oi=ao.

Awards and Fellowships

FAPEPS Fellowship Stipend — **07/2026 - current**

Best Master's Dissertation at SIBGRAPI — **10/2026**

PhD CAPES Foundation Fellowship Stipend — **11/2025 - 06/2026**

Master's CAPES Foundation Fellowship Stipend — **2023-2025**

Ministry of Education: Educational Tutoring Program (PET) Fellowship Stipend — **2019-2022**

Covid-19 spreading prediction in Paraná state. — **2021**
Honorable Mention: 6º Congresso Paranaense de Saúde Púbica/Coletiva

Teaching activities

LaTeXfor beginners week course — **2021 & 2022**
Department of Mathematics State University of Londrina

Teaching Assistant for Ordinary Differential Equations and Basic Mathematics — **2019 & 2020**
Department of Mathematics State University of Londrina

Teaching Assistant for Laboratory of Advanced Algorithm (1 and 2) — 2024/1 & 2024/2
Computer Science ICMC - USP - Universidade de São Paulo

Teaching assistant for 3D Reconstruction — 2025/2
IMPA - Instituto de Matemática Pura e Aplicada

Event organization: XXXIV UEL Mathematics Week Conference 2022; Basic Maths Week 2021; XIV JOPARPET 2020; 10ª Feira Das Profissões: Conhecendo A Uel. 2022

Website development: XXXIV UEL Mathematics Week Conference 2022; GIVA Labs

Programming Skills

Python: Pandas, Plotly, Sklearn, Pytorch, OpenCV, Pyside6 (Qt), and others

Miscellaneous: LaTeX, Gnuplot, HTML, CSS, Fortran, C#, SQL, JavaScript

Languages

Português: Native

English: highly-profficient

Français: 2 semesters

Contact

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Linkedin

Curriculum Lattes

Scholar